

Assignment Structure and Expectations

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Assignment Title: - Research and Present a Sorting Algorithm (to be done in Groups)

Objective:

Understand, analyze, and effectively communicate the working of a sorting algorithm – both theoretically and practically – with a focus on clarity, comparison, and real-world relevance.

Instructions:

Each group has been assigned a sorting algorithm. You are required to research, analyze, and present your findings through a well-structured slide deck and a short demo.

Your presentation must include:

- 1. Overview of the Algorithm**

- What it is and where it fits in the sorting family.

- 2. Step-by-step Explanation**

- Use a **diagram or visual flow** to explain how it works on a sample list of numbers.

- 3. Sample Code**

- Provide a clean and well-commented implementation in your preferred language (Python/C++/Java, etc.)
- Include walkthrough of the code logic during the presentation.

- 4. Complexity Analysis**

- Time Complexity: Best, Average, Worst case

- Space Complexity
- Stability (is the algorithm stable?)

5. **Comparison**

- Briefly compare your algorithm with the closest related one in terms of approach or performance. (e.g., Bubble Sort vs Insertion Sort)

6. **Applications**

- Where is your algorithm used in the real world or in specific problem scenarios?

7. **Strengths and Limitations**

- When is it ideal to use this algorithm, and when is it not?

Deliverables:

- A slide deck (PDF or PowerPoint)
- Code implementation (submitted separately or in a shared GitHub repo)
- 7-10 minute in-class presentation

Deadline:

[Presentations will be made during our Virtual class sessions next week]

Grading Criteria:

- Clarity and structure of slides
- Depth of explanation
- Accuracy of implementation

- Understanding of complexities and comparisons
 - Presentation and communication skills
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